

Practice Quiz: Cognition

Name: __ Date: __

Part A: Multiple Choice

1. Variational free energy is best understood as: A) The energy consumed by the brain B) A measure of how well the agent's model fits reality — lower is better C) The physical heat generated by neural computation D) A reward signal that drives behavior
2. The accuracy-complexity trade-off means the brain: A) Always prefers the most accurate model regardless of complexity B) Always prefers the simplest model regardless of accuracy C) Balances fitting the data well with keeping the model as simple as possible D) Never changes its model
3. Attention in Active Inference is: A) A limited resource that gets used up B) The dynamic adjustment of precision — controlling how much weight different information receives C) The same as consciousness D) Only related to visual processing
4. The Dark Room problem asks why agents don't: A) Sleep all the time B) Seek out the most predictable, boring environment to minimize surprise C) Eat more food D) Pay attention to everything equally
5. Confirmation bias can be explained by Active Inference as: A) Random errors in thinking B) High prior precision that downweights disconfirming sensory evidence C) A lack of intelligence D) Insufficient memory capacity
6. Which neurotransmitter is associated with boosting sensory precision ("pay attention to what you're sensing")? A) Dopamine B) Serotonin C) Acetylcholine D) Endorphin
7. In the Active Inference account, delusions may arise from: A) Refusing to think about reality B) Excessively high precision on aberrant prediction errors, making false beliefs feel certain C) Low intelligence D) Too much sleep

Part B: Short Answer

1. A friend tells you "I just can't stop worrying about everything." Using Active Inference concepts (precision, prediction error, priors), explain what might be happening in your friend's brain and why they find it hard to stop.
2. Explain the Dark Room problem and its solution. Why don't agents just minimize surprise by staying in a completely predictable environment? Reference the role of preferences and the generative model.
3. Compare two students: Student A studies a subject they already believe they understand well. Student B studies a subject they know very little about. Using the accuracy-complexity trade-off, explain which student is likely to experience more free energy, and why.